MILITARY SPECIFICATION

DUPLICATING MACHINE, STENCIL-PROCESS, TABLE MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two styles of table mounted, stencil-process, duplicating machines.

1.2 Classification. The duplicating machines shall be of the following styles as specified (see 6.2).

Style 1 - With separate storage cabinet
Style 2 - Without separate storage cabinet

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-535	-	Ink, Duplicating Machine Stencil, Paste, Black
UU-P-388	-	Paper, Mimeograph
UU-S-1205	-	Stencil, Duplicating Machine
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-576	-	Boxes, Wood, Cleated, Veener, Paper Overlaid
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command (GL), Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Set-Up
PPP-C-843 - Cushioning Material, Cellulosic
PPP-P-291 - Paperboard, Wrapping and Cushioning

MILITARY

MIL-P-116 - Preservation-Packaging, Methods Of
MIL-B-121 - Barrier-Material, Greaseproofed, Water-Proofed, Flexible
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirement for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of

(Copies of specifications, standards, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS LABORATORIES INC. (UL) STANDARD

UL 114 - Standard for Office Appliances and Business Equipment

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, Long Island, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; 2602 Tampa East Blvd., Tampa, FL 33619.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. The machines furnished under this specification shall be the manufacturer's current standard product except for changes necessary to comply with the requirements of this specification.

3.2 UL compliance and certification. The units shall comply with the applicable requirements of UL 114. Prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the machines furnished under this specification meet the applicable requirements of UL. Acceptable evidence of meeting the requirements shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the machine has been tested and conforms to UL 114.

3.3 Materials and components. The materials and components shall be as specified herein. Materials and components not specifically specified shall be of the quality normally used by the manufacturer in his standard machine provided the completed item complies with all the provisions of this specification (see 6.3).

3.4 Design and construction. Unless otherwise specified (see 6.2), the machine shall operate on a nominal 120 volts, 60 hertz, single phase. The machine shall automatically feed, ink, print and deliver paper. The machine shall be provided with a cylinder designed for paste ink. Automatic ink distribution to the printing surface shall be provided. The feed and receiving trays' capacity shall be a minimum of two hundred and fifty sheets of paper. The area for duplicating copy shall not be less than 7 x 13-1/2 inches (178 x 343 mm). Means shall be provided for adjusting the duplicating area in relation to the paper edges. Means for disengaging the

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impression roller and feed mechanism shall be provided. Copy counters shall be visible to the operator and have a maximum setting of not less than 999 copies. A control shall be located on the surface of the variable speed mechanism for adjusting the duplicating speeds. The power unit shall be built within the duplicator and shall be readily accessible for repair and maintenance. Water repellent and readily removable dust covers shall be furnished with each machine. A clamp for attaching the stencil shall be provided. Style 1 duplicator shall be provided with a metal storage cabinet. The cabinet shall be provided with magnetic doors and compartments for storage with a minimum of one shelf. The storage cabinet legs shall be provided with glides and shall have a vertical adjustment of not less than 1 inch (25.4 mm). Style 2 duplicator shall be provided without a storage cabinet.

3.5 Performance.

3.5.1 Continuous speed. The unit shall run continuously for 1 hour with a speed not less than 125 copies per minute, without damage or malfunction, when tested as specified in 4.3.1.

3.5.2 Ink distribution. The unit shall provide printed copies with black squares that comply with an ink distribution value of not less than 0.70, when the duplicator is tested as specified in 4.3.2.

3.5.3 Registration. The unit shall provide printed copies with lines that do not vary more than 1/32 inch (.8 mm) from the left and top edges of the paper, when the duplicator is tested as specified in 4.3.3.

3.5.4 Lateral and vertical adjustment. The unit shall provide printed copies where the differences between the margins of the top and left edges on two papers shall not be less than 1/2 and 1-3/4 inches (12.7 and 44.5 mm) respectively, when the duplicator is tested as specified in 4.3.4.

3.5.5 Feed mechanism. The feed mechanism shall operate without jamming or misfeeding of the paper, when the duplicator is tested as specified in 4.3.5.

3.5.6 Electromagnetic compatibility. When specified (see 6.2), the unit shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.3.6).

3.6 Finish. The units shall be finished in accordance with the manufacturer's commercial practice.

3.7 Marking. The units shall be marked for identification in accordance with the manufacturer's commercial practice and shall include manufacturer's name, date of manufacture, model number, and electrical characteristics.

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3.8 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.9 Workmanship. The units shall be free from defects such as fractures, splits, punctures, tears, dents, creases, deterioration, or malformation. There shall be no sharp edges, slivers, burrs or projections. Threaded fasteners shall not be broken, stripped, fractured or loose. Rivets shall completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members and shall not be loose.

3.9.1 Application of finish. The finish applied to the unit shall be continuous, smooth, adherent, have no imbedded foreign matter, and contain no sags, runs, drips, creeps, laps, bubbles or streaks. No rust or tool marks shall show through the coating.

3.9.2 Welding. The surfaces of parts to be welded shall be free from oxide, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux shall be removed from the finished welds. Soldering shall be complete and adherent, with all flux and residue removed and shall contain no pinholes.

3.9.3 Electrical wiring. Wiring shall not be cut, abraded, or have insulation stripped, and shall be joined at the terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling shall be performed in accordance with MIL-STD-105.

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4.2.1 Component and material inspection. In accordance with 4.1, the components and materials shall be inspected in accordance with all the requirements of the referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification.

4.2.2 End item inspection. The lot shall be all duplicating machines offered for inspection at one time. The sample unit for this examination shall be one assembled duplicating machine.

4.2.2.1 Visual examination. The completely assembled item shall be examined for defects in table I. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 for major and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Finish not in accordance with the manufacturer's commercial practice.	X	
	Not continuous, smooth, adherent.	X	
	Foreign matter, imbedded in finish.		X
	Sags, runs, drips, creeps, laps, bubbles or streaks.		X
	Rust or tool marks.	X	
Design and construction and workmanship, general (applicable to all components and assemblies)	Duplicator not style required.	X	
	Any component or part missing or not specific type or size.	X	
	Fractured, split, torn, dented, creased, deteriorated or malformed.	X	
	Component misplaced, or not in proper alignment.		X
	Component loose.		X
	Sharp burrs, slivers or sharp edges.		X
	No automatic ink distribution to the printing surface.	X	
	No means for adjusting the duplicating area in relation to the paper edges.	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Design and construction and workmanship, general (applicable to all components and assemblies) (cont'd)	No means for disengaging the impression roller and feed mechanism.	X	
	Copy counter not visible to operator.		X
	Control for variable speed mechanism not located on the surface.	X	
	Power unit not built within the duplicator.	X	
	Power unit not readily removable.	X	
	Protective covers not readily removable.		X
Welding	Missing, incomplete, burn holes, cracked, fractured, not fused.	X	
	Undercut not smooth and uniform or scale or flux deposits not removed.		X
Soldering (when applicable)	Missing, not adherent or incomplete.		X
	Not clean (flux residue not removed, pinholes in solder).		X
Bolts, nuts, screws, studs and other types of threaded fasteners	Broken, stripped, fractured or not tight.		X
Rivets (when required)	Broken, not peened or insufficiently peened or loose.	X	
Electrical assembly	Wiring cut, abraded, or not properly installed or joined.	X	
	Adequate slack not provided for wiring to relieve strain.		X
Marking	Missing, incomplete, not legible, not specified data.		X

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4.2.2.2 Dimensional examination. Examination shall be made of the duplicating machine for compliance with dimensions specified. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units. Any nonconformance shall be classified a defect.

4.2.2.3 Testing of the end item. When electromagnetic compatibility is required the first produced end item shall be as specified in 4.3.6. In addition, each end item shall be tested as specified in 4.3.1 through 4.3.5. Failure to pass either test shall be cause to reject the effected end item.

4.2.3 Packaging inspection. Packaging inspection shall be made to determine that preservation-packaging, packing and marking comply with section 5. Defects shall be scored as set forth in table II. The sample unit shall be one shipping container fully packed, except that it need not be closed. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-1 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II. Packaging inspection

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size; location, sequence, or method of application.
Preservation	Improperly applied or missing.
Materials	Components missing, damaged.
Workmanship	Inadequate application of components such as incomplete closure of barrier materials not taped, as specified.

4.3 Tests. The mimeograph paper for tests shall be white, 8-1/2 x 14 inches (216 x 355.6 mm) and shall conform to types I and III of UU-P-388. Type I paper shall be used in tests 4.3.2, 4.3.4 and 4.3.5. Type III paper shall be used in test 4.3.3. The paper stencils for tests shall be 8-1/2 x 14 inches (216 x 355.6 mm) and shall conform to the requirements of UU-S-1205. The stencil used in test 4.3.2 shall be marked with nine horizontal

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and five vertical rows of 3/8 inch (9.5 mm) black squares with equal centered distances and 6-3/4 x 13-1/8 inches (171.5 x 333.4 mm) outer boundaries. The stencils used in tests 4.3.3, 4.3.4 and 4.3.5 shall be marked with two lines each equidistant by 1/8 inch (3.2 mm) from the left and top edges. The ink for the applicable tests 4.3.2, 4.3.3, 4.3.4 and 4.3.5 shall conform to composition A of TT-I-535.

4.3.1 Continuous speed test. Without paper and ink, the duplicator shall be operated at a speed not less than 125 copies per minute for 1 hour. Any evidence of damage or malfunction shall constitute failure of this test (see 3.5.1).

4.3.2 Ink distribution test. Using the specified stencil and paper, 100 copies shall be run off. Densitometer readings shall be taken of each black square on each of the final 3 copies duplicated, and the ink distribution value determined. If the ration of the minimum average density to the maximum average density of each horizontal and vertical row is less than 0.70, the machine shall have failed to pass this test (see 3.5.2).

4.3.3 Registration test. Using the specified stencil and paper, 100 copies shall be run off. If on the first and last copies duplicated, the distance of each of the two lines from the corresponding left and top edge of the paper varies more than 1/32 (.8 mm), the machine shall have failed to pass this test (see 3.5.3).

4.3.4 Lateral and vertical adjustment test. Using the specified stencil and paper, 1 copy shall be run off with the machine set to give a maximum margin at the left and top edges. With the machine set to a minimum margin another copy shall be run off. If the differences between the widths of the left and top margins on the two sheets is less than 1/2 and 1-3/4 inches (12.7 x 44.5 mm) respectively, the machine shall have failed to pass this test (see 3.5.4).

4.3.5 Feed mechanism test. If the machine does not feed or misfeeds more than once, the machine shall have failed to pass this test (see 3.5.5).

4.3.6 Electromagnetic compatibility test. When electromagnetic compatibility test is required, the initial production units, as applicable, shall be tested by the contractor in accordance with Test Method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461 (see 3.5.6).

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Protruding and detachable components shall be removed from each unit to reduce the cube of the pack or to prevent damage to the component.

5.1.1.2 Cleaning and drying. All exposed, uncoated ferrous metal surfaces of each unit shall be cleaned in accordance with process C-1 of MIL-P-116 and thoroughly dried.

5.1.1.3 Application of preservative. Surfaces, cleaned and dried as specified, shall be coated with type P-7 preservative of MIL-P-116.

5.1.1.4 Protection of preservative. Surfaces from which a preservative might be removed by contact with preservation and packing materials, shall be covered or wrapped with greaseproof barrier material conforming to type I or II, grade A of MIL-B-121. Barrier material shall be secured with twine or with tape, conforming to PPP-T-45, type III, grade A or heat sealed when heat-sealable barrier material is used.

5.1.1.5 Securing. Parts of the machine that are free to move shall be secured to prevent movement or vibration while in transit.

5.1.1.6 Preservation. Detached components shall be wrapped in cushioning material conforming to PPP-C-843 or PPP-P-291, or preserved in a folding or set-up paperboard box or a fiberboard box conforming to PPP-B-566, PPP-B-676, or type I or II, class 1 of PPP-B-636, respectively. Cushioning material shall be secured with tape, cord or twine. Each box shall be securely closed in accordance with the appendix of the box specification.

5.1.2 Level C. Units shall be preserved-packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A. Each unit, including detached components, and dust covers, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting shipping container, conforming to class WR, V3c or V3s, style RSC of PPP-B-636; overseas type of PPP-B-591 or PPP-B-601; class 2 of PPP-B-576; or class 2, style 2 or 4 of PPP-B-621. The weight and dimensional of the applicable container specification shall not be exceeded. The contents shall be immobilized within the shipping container by means of fiberboard inserts. Cushioning material shall be used, as necessary, to protect the unit while in transit. Each fiberboard shipping container shall be waterproofed by means of tape or a case liner, in accordance with the appendix of the container specification. All other shipping containers shall be provided with a type I or II, grade B or C, sealed case liner, conforming to MIL-L-10547. Each shipping container shall be reinforced with flat steel strapping or the tape banding in accordance with the appendix of the applicable container specification.

5.2.2 Level B. Each unit, including detached components, tools, and dust cover, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting shipping container, conforming to type CF or SF, class WR, style RSC of PPP-B-636; domestic type, class 1, style A or B of PPP-B-591 or PPP-B-601; class 1 of PPP-B-576, or class 1, style 2 or 4 of PPP-B-621. The contents shall be immobilized within the shipping container by means of fiberboard inserts or other means. Cushioning material shall be used, as necessary, to protect the unit while in transit. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps or by a combination of metal, stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 Level C. Units preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for use in reproducing on paper markings from a prepared stencil.

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6.2 Ordering data. Procurement data should specify the following:

(a) Title, number, and date of this specification.
(b) Style of duplicating machine required (see 1.2).
(c) Electrical characteristics other than specified (see 3.4).
(d) When electromagnetic compatibility is required (see 3.5.6).
(e) Contract data requirements (see 3.8).
(f) Applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by the specification shall be specifically called for in the contract in accordance with the applicable regulations of the procuring agency (see 3.8 and 6.2).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.4 Metric equivalents. Metric equivalents indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

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Navy - SA
Air Force - 99

Preparing activity:

Army - GL

Project No. 3610-0282

Review activity:

DLA - GS

User activity:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-43971 - Duplicating Machine, Stencil-Process, Table Mounted

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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